

1.4: Transformations

$$y = a[k(x-d)]^n + c$$

- $a \Rightarrow$ vertical stretch or compression
if $a < 0$ there will be a reflection in the x-axis
- $k \Rightarrow$ horizontal stretch or compression by $1/k$
if $k < 0$ there will be a reflection in the y-axis
- $c \Rightarrow$ vertical translation up or down
- $d \Rightarrow$ horizontal translation to the left or right

The polynomial function $y = a[k(x - d)]^n + c$ can be graphed by applying the above transformations (with parameters a, k, d and c) to the graph of the base or parent function $y = x^n$.

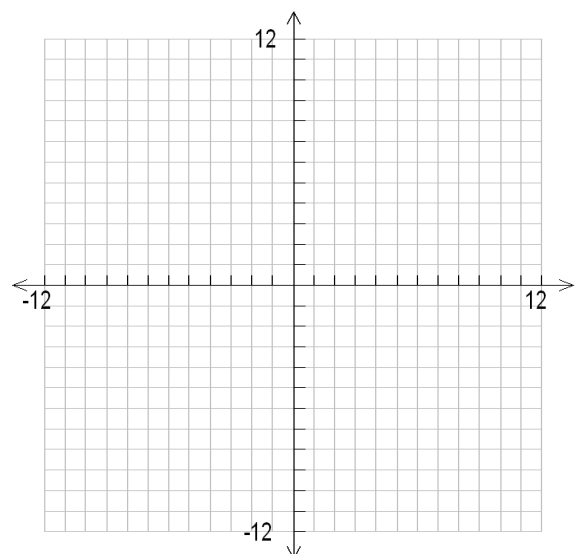
Mapping Rule: $(\underline{x}, \underline{y}) \rightarrow (\frac{1}{k}x + d, ay + c)$

Example 1:

Given the transformation $y = -8(\frac{1}{2}x + 2)^3 - 3$.

a) Describe the transformations applied to the base graph $y = x^3$.

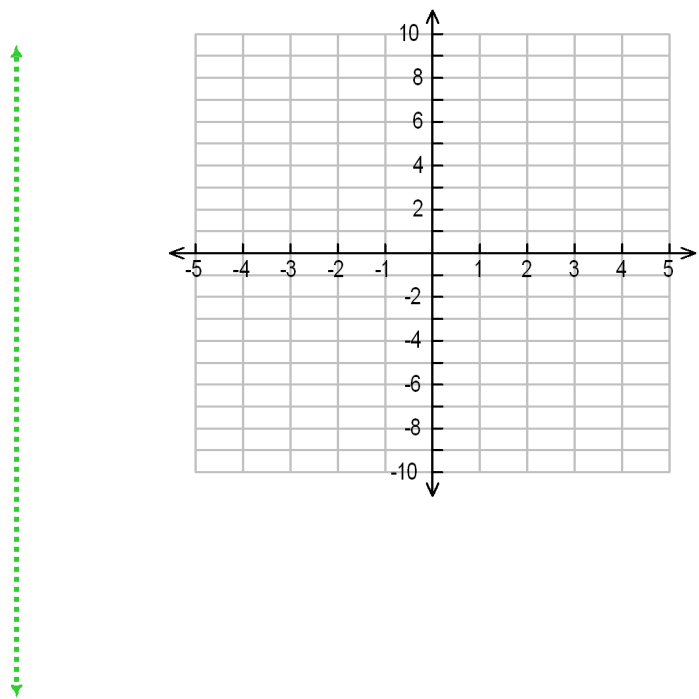
b) Using a mapping rule, sketch the base graph and the transformed graph.



Example 2:

Given $f(x) = \frac{1}{2}(x + 1)^4 - 3$. Sketch $f(x)$.

State the domain, range and axis of symmetry.



$(-2, 16)$	$(-3, 5)$
$(-1, 1)$	$(-2, -2.5)$
$(0, 0)$	$(-1, -3)$
$(1, 1)$	$(0, -2.5)$
$(2, 16)$	$(1, 5)$

Note:

When n is even, the graph of the polynomial function $y=a[k(x-d)]^n + c$ will have vertex at (d,c) and will have line symmetry (*but not in the y axis*). The axis of symmetry is $x = d$.

When n is odd, the graph of the polynomial function $y=a[k(x-d)]^n + c$ will have rotational (point) symmetry (*but not about the origin*), about the point (d, c) .

WHY? Because these are transformed power functions (which always have symmetry).

* Even / Odd FUNCTION vs. Even / Odd DEGREE *

HOMEWORK

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